

MA1002 — Calculus for Science Part 2

Summer 2022 Examination · Paper 1 · Full Marking Scheme

Module	MA1002 — Calculus for Science Part 2
Internal examiner	Ms. P. Chaudhry
External examiner	Prof. G. Ellis
Head of School	Dr. K. Hayes
Institution	University College Cork
Session	Summer 2022, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer 4 of the 5 questions . All questions carry equal marks (17.5 each, 70 in total). Marks for each part are shown in brackets.
Materials	No calculators. Only the log tables and formula sheet provided may be used.

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Coverage note. On the day you answer 4 questions, but this scheme works through **all five** in full, so you can pick the ones you’re strongest on.

Source basis. Every answer uses MA1002 course methods: indefinite and definite integrals, substitution, integration by parts, area, volumes of rotation, partial fractions and separable differential equations. Two parts go outside the listed topics and carry a note: Q4(b) (velocity and position) and Q5(a) (regression line, formulas supplied).

Mark splits. Part totals are the paper’s own. The step-by-step split within each part is Study-Uni’s suggested allocation.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

No calculators. Every answer is exact (e.g. $\frac{4}{3}$, $\frac{\pi}{5}$, $\frac{59}{3}$) and all the arithmetic is shown by hand.

Question 1

Antiderivatives (the indefinite integral), initial conditions — 17.5 marks

1(a)

[8 marks]

Find all the functions f such that $f'(x) = x^4 + 3 \cos 3x + e^{5x} + \frac{5}{x} - \frac{x^3 - \sqrt{x}}{x} + 7$.

ANSWER

$$f(x) = \frac{x^5}{5} + \sin 3x + \frac{1}{5}e^{5x} + 5 \ln|x| - \frac{x^3}{3} + 2\sqrt{x} + 7x + C$$

RULES YOU NEED

- $\int \cos(ax) \, dx = (1/a) \sin(ax)$, and $\int e^{ax} \, dx = (1/a) e^{ax}$.
- $\int x^n \, dx = x^{n+1}/(n+1)$ ($n \neq -1$), and $\int (1/x) \, dx = \ln|x|$.
- $\sqrt{x} = x^{1/2}$, and $x^{1/2} \div x = x^{-1/2}$.
- "All the functions" means add $+ C$.

WORKING — STEP BY STEP

1 Split the fraction

Divide each term on top by x . The minus sign in front goes onto both terms:

$$\frac{x^3 - \sqrt{x}}{x} = x^2 - x^{-1/2} \Rightarrow -\frac{x^3 - \sqrt{x}}{x} = -x^2 + x^{-1/2}$$

$$f'(x) = x^4 + 3 \cos 3x + e^{5x} + \frac{5}{x} - x^2 + x^{-1/2} + 7$$

2 Integrate the cos and exponential terms

Here $a = 3$ for $\cos$ and $a = 5$ for e :

$$3 \cos 3x \rightarrow 3 \times \frac{1}{3} \sin 3x = \sin 3x$$

$$e^{5x} \rightarrow \frac{1}{5} e^{5x}$$

3 Integrate $5/x$

$$5 \cdot (1/x) \rightarrow 5 \ln|x|$$

4 Integrate the powers and the constant

Add 1 to the power, divide by the new power ($-1/2 + 1 = 1/2$, and $\div 1/2$ means $\times 2$):

$$x^4 \rightarrow \frac{x^5}{5} \quad -x^2 \rightarrow -\frac{x^3}{3} \quad x^{-1/2} \rightarrow \frac{x^{1/2}}{1/2} = 2\sqrt{x} \quad 7 \rightarrow 7x$$

5 Put it together, + C

$$f(x) = \frac{x^5}{5} + \sin 3x + \frac{1}{5}e^{5x} + 5 \ln|x| - \frac{x^3}{3} + 2\sqrt{x} + 7x + C$$

EXAM TECHNIQUE

The minus in front of the fraction applies to both terms: $-(x^2 - x^{-1/2}) = -x^2 + x^{-1/2}$.

Marks	Awarded for
1.5	Fraction split: $-x^2 + x^{-1/2}$.
1	$x^4 \rightarrow x^5/5$.
1	$3 \cos 3x \rightarrow \sin 3x$.
1	$e^{5x} \rightarrow (1/5)e^{5x}$.
1	$5/x \rightarrow 5 \ln x $.
1.5	$-x^3/3 + 2\sqrt{x}$.
1	$7x$ and $+ C$.

1(b)**[9.5 marks]**Find g given that $g''(x) = 24x^2 + 2x + 10$, $g(1) = 5$ and $g'(1) = -3$.**ANSWER**

$$g(x) = 2x^4 + \frac{1}{3}x^3 + 5x^2 - 22x + \frac{59}{3}$$

RULES YOU NEED

- $\int x^n dx = x^{n+1}/(n+1)$.
- Integrate g'' to get g' (constant C), then g' to get g (constant D).
- $g'(1) = -3$ finds C ; $g(1) = 5$ finds D .

WORKING — STEP BY STEP**1 Integrate $g''(x)$ to get $g'(x)$**

$$24x^2 \rightarrow 24 \cdot \frac{x^3}{3} = 8x^3 \quad 2x \rightarrow x^2 \quad 10 \rightarrow 10x$$

$$g'(x) = 8x^3 + x^2 + 10x + C$$

2 Use $g'(1) = -3$

$$8 + 1 + 10 + C = -3 \Rightarrow 19 + C = -3 \Rightarrow C = -22$$

3 Integrate $g'(x)$ to get $g(x)$

$$8x^3 \rightarrow 2x^4 \quad x^2 \rightarrow \frac{x^3}{3} \quad 10x \rightarrow 5x^2 \quad -22 \rightarrow -22x$$

$$g(x) = 2x^4 + \frac{x^3}{3} + 5x^2 - 22x + D$$

4 Use $g(1) = 5$ Put $x = 1$ in, then write everything in thirds:

$$2 + \frac{1}{3} + 5 - 22 + D = 5 \Rightarrow -15 + \frac{1}{3} + D = 5$$

$$-\frac{44}{3} + D = 5 \Rightarrow D = \frac{15}{3} + \frac{44}{3} = \frac{59}{3}$$

5 Check

$$g(1) = 2 + \frac{1}{3} + 5 - 22 + \frac{59}{3} = -15 + \frac{60}{3} = -15 + 20 = 5 \checkmark$$

EXAM TECHNIQUE

No calculator: turn whole numbers into thirds ($-15 = -45/3$) so the fractions add cleanly.

Marks	Awarded for
2	$g'(x) = 8x^3 + x^2 + 10x + C.$
2	$C = -22.$
2.5	$g(x) = 2x^4 + x^3/3 + 5x^2 - 22x + D.$
2	$D = 59/3.$
1	Final $g(x)$ stated.

Question 1 Total: 17.5 Marks

Question 2

Substitution, area between curves, definite vs indefinite integrals — 17.5 marks

2(a)

[7 marks]

Use Integration by Substitution to evaluate $\int \sin^2 x \cos x \, dx$.

ANSWER

$$\frac{\sin^3 x}{3} + C$$

RULES YOU NEED

- Substitution: let $u = \sin x$, so $du = \cos x \, dx$.
- $\int u^2 \, du = u^3/3$.

WORKING — STEP BY STEP

1 Choose u

The derivative of $\sin x$ is $\cos x$, and $\cos x$ is sitting outside:

$$u = \sin x \Rightarrow du = \cos x \, dx$$

2 Substitute

$\sin^2 x$ becomes u^2 , and $\cos x \, dx$ becomes du :

$$\int \sin^2 x \cos x \, dx = \int u^2 \, du$$

3 Integrate

$$\frac{u^3}{3} + C$$

4 Put x back in

$$\frac{\sin^3 x}{3} + C$$

EXAM TECHNIQUE

$\sin^2 x$ means $(\sin x)^2$, so with $u = \sin x$ it is simply u^2 .

Marks	Awarded for
2.5	$u = \sin x$, $du = \cos x \, dx$.
1.5	Integral becomes $\int u^2 \, du$.
2	Integrated: $u^3/3$.
1	$\sin^3 x/3 + C$.

2(b)**[7 marks]**

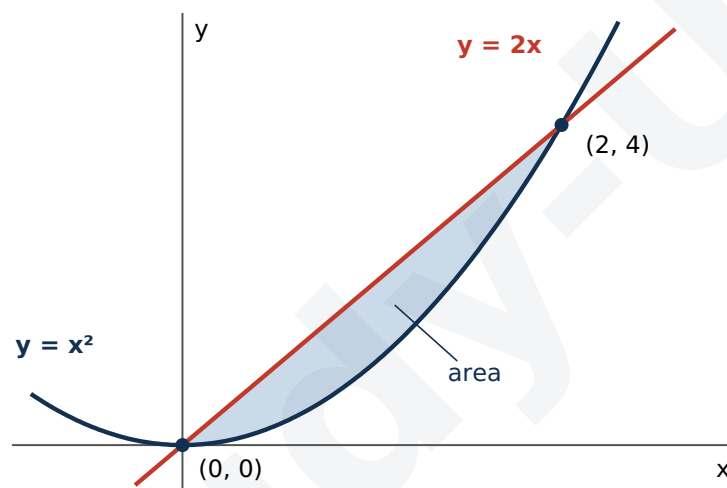
Find the area enclosed by the parabola $y = x^2$ and the line $y = 2x$. Sketch this area.

ANSWER

$$\text{Area} = \frac{4}{3} \text{ square units}$$

RULES YOU NEED

- Area between two curves = $\int (\text{upper curve} - \text{lower curve}) dx$.
- The limits are the x-values where the curves meet.



Sketch: the line $y = 2x$ is above the parabola between $x = 0$ and $x = 2$. The shaded region is the area.

WORKING — STEP BY STEP**1 Find where they meet**

$$x^2 = 2x \Rightarrow x^2 - 2x = 0 \Rightarrow x(x - 2) = 0 \Rightarrow x = 0 \text{ or } x = 2$$

2 Which is on top?

Test $x = 1$:

$$\text{Line: } 2(1) = 2 \quad \text{Parabola: } 1^2 = 1 \Rightarrow \text{line on top}$$

3 Set up and integrate

$$A = \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2$$

4 Put in the limits

$$4 - \frac{8}{3} = \frac{12}{3} - \frac{8}{3} = \frac{4}{3}$$

5 Sketch

Draw the parabola through (0, 0), the line through the origin, mark where they cross at (0, 0) and (2, 4), and shade between them (see the diagram above).

EXAM TECHNIQUE

The question says “Sketch”, so the sketch carries a mark. Label both curves and both crossing points.

Marks	Awarded for
1.5	Intersections $x = 0$ and $x = 2$.
1	Line identified as upper curve.
1.5	$A = \int_0^2 (2x - x^2) dx$.
1	Integrated: $x^2 - x^3/3$.
1	$A = 4/3$.
1	Sketch with curves, crossing points and region shown.

2(c)**[3.5 marks]**

Explain the difference between a definite integral and an indefinite integral.

ANSWER

Indefinite $\int f(x) dx = F(x) + C$ is a family of functions.

Definite $\int_a^b f(x) dx = F(b) - F(a)$ is a single number.

RULES YOU NEED

- An antiderivative F of f is a function with $F'(x) = f(x)$.
- Fundamental Theorem: $\int_a^b f(x) dx = F(b) - F(a)$.

WORKING — STEP BY STEP**1 Indefinite integral**

Has no limits. It gives **every** antiderivative of f , so the answer is a function plus a constant:

$$\int f(x) dx = F(x) + C, \text{ where } F'(x) = f(x)$$

Example: $\int 2x dx = x^2 + C$.

2 Definite integral

Has limits a and b . It gives a single **number**: the signed area under $y = f(x)$ from $x = a$ to $x = b$:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Example: $\int_0^1 2x dx = 1^2 - 0^2 = 1$.

3 The link

You work out a definite integral by finding an indefinite one and putting in the limits. The $+ C$ cancels when you subtract, so a definite integral has no $+ C$.

Marks	Awarded for
1.5	Indefinite: no limits, family of antiderivatives $F(x) + C$ (a function).
1.5	Definite: limits a to b , a number $F(b) - F(a)$ (area).
0.5	Link: C cancels / Fundamental Theorem.

Question 2 Total: 17.5 Marks

Question 3

Partial fractions, integration by parts — 17.5 marks

3(a)

[8 marks]

Find $\int \frac{x+7}{(x+1)(x+3)} dx$ by using the method of partial fractions.

ANSWER

$$3 \ln|x+1| - 2 \ln|x+3| + C$$

RULES YOU NEED

- Write the fraction as $A/(x+1) + B/(x+3)$.
- Find A and B by putting in the x-values that make each bracket zero.
- $\int 1/(x+a) dx = \ln|x+a|$.

WORKING — STEP BY STEP

1 Set up the partial fractions

Multiply both sides by $(x+1)(x+3)$:

$$\frac{x+7}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3} \Rightarrow x+7 = A(x+3) + B(x+1)$$

2 Find A: put $x = -1$

$$-1+7 = A(2) \Rightarrow 6 = 2A \Rightarrow A = 3$$

3 Find B: put $x = -3$

$$-3+7 = B(-2) \Rightarrow 4 = -2B \Rightarrow B = -2$$

Quick check: $A + B = 1$, the coefficient of x on top ✓.

4 Integrate each fraction

$$\int \left(\frac{3}{x+1} - \frac{2}{x+3} \right) dx = 3 \ln|x+1| - 2 \ln|x+3| + C$$

EXAM TECHNIQUE

Use the x-values that make a bracket zero: each one removes a term and gives A or B in one line.

Marks	Awarded for
2	$x + 7 = A(x + 3) + B(x + 1)$ set up.
2	$A = 3$.
2	$B = -2$.
2	$3 \ln x + 1 - 2 \ln x + 3 + C$.

3(b)**[9.5 marks]**Use Integration by Parts to find $\int x^2 \ln x \, dx$.**ANSWER**

$$\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$$

RULES YOU NEED

- Integration by parts: $\int u \, v' \, dx = u \, v - \int u' \, v \, dx$.
- With $\ln x$ in the integral, let $u = \ln x$.
- $d/dx (\ln x) = 1/x$.

WORKING — STEP BY STEP**1 Choose u and v'**

$$u = \ln x \Rightarrow u' = \frac{1}{x} \quad v' = x^2 \Rightarrow v = \frac{x^3}{3}$$

2 Apply the parts formula

$$\int x^2 \ln x \, dx = \frac{x^3}{3} \ln x - \int \frac{1}{x} \cdot \frac{x^3}{3} \, dx$$

3 Simplify and integrate

$$(1/x)(x^3/3) = x^2/3:$$

$$\int \frac{x^2}{3} \, dx = \frac{1}{3} \cdot \frac{x^3}{3} = \frac{x^3}{9}$$

4 Put it together

$$\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$$

EXAM TECHNIQUE

Don't choose $u = x^2$: you'd need $v = \int \ln x \, dx$, which is harder. $u = \ln x$ makes the second integral easy.

Marks	Awarded for
3	$u = \ln x$, $v = x^3/3$ (correct choice).
3	Parts formula applied correctly.
2	$\int x^2/3 \, dx = x^3/9$.
1.5	Final answer, + C.

Question 3 Total: 17.5 Marks

Question 4

Separable differential equations, velocity and position — 17.5 marks

4(a)

[8 marks]

Use separation of variables to solve the differential equation $\frac{dy}{dx} = e^x \cos^2 y$ subject to the initial condition $y(0) = \frac{\pi}{4}$.

ANSWER

$$y = \tan^{-1}(e^x)$$

RULES YOU NEED

- Separation of variables: y terms with dy on one side, x terms with dx on the other, then integrate.
- $1/\cos^2 y = \sec^2 y$, and $\int \sec^2 y \, dy = \tan y$ (since $d/dy \tan y = \sec^2 y$).
- $\tan(\pi/4) = 1$, and $e^0 = 1$.

WORKING — STEP BY STEP

1 Separate the variables

Divide by $\cos^2 y$, multiply by dx:

$$\frac{1}{\cos^2 y} \, dy = e^x \, dx \Rightarrow \sec^2 y \, dy = e^x \, dx$$

2 Integrate both sides

$$\int \sec^2 y \, dy = \int e^x \, dx \Rightarrow \tan y = e^x + C$$

3 Use $y(0) = \pi/4$

$$\tan(\pi/4) = e^0 + C \Rightarrow 1 = 1 + C \Rightarrow C = 0$$

$$\tan y = e^x$$

4 Solve for y

Take the inverse tan of both sides:

$$y = \tan^{-1}(e^x)$$

5 Check

$$y(0) = \tan^{-1}(1) = \pi/4 \checkmark$$

EXAM TECHNIQUE

Recognise $1/\cos^2 y$ as $\sec^2 y$. It integrates straight to $\tan y$ (the formula sheet has $d/dx \tan x = \sec^2 x$).

Marks	Awarded for
2	Separated: $\sec^2 y \, dy = e^x \, dx$.
2.5	Integrated: $\tan y = e^x + C$.
2	$C = 0$ from $y(0) = \pi/4$.
1.5	$y = \tan^{-1}(e^x)$.

4(b)**[9.5 marks]**

A particle moves in a straight line. Its velocity at time t is $v(t) = t^2 - t - 6$ cm/s. Find the position of the particle after 6 seconds if its initial displacement is $s(0) = 9$ cm.

ANSWER

$$s(6) = 27 \text{ cm}$$

NOTE

The link between position and velocity ($v = ds/dt$) comes from MA1001. After that, this is an antiderivative with an initial condition, as in Question 1.

RULES YOU NEED

- Velocity is the derivative of position: $v = ds/dt$, so $s(t) = \int v(t) dt$.
- $\int t^n dt = t^{n+1}/(n+1)$.
- Use $s(0) = 9$ to find the constant.

WORKING — STEP BY STEP**1 Integrate velocity to get position**

$$s(t) = \int (t^2 - t - 6) dt = \frac{t^3}{3} - \frac{t^2}{2} - 6t + C$$

2 Use $s(0) = 9$

Every t term is 0 at $t = 0$:

$$s(0) = C = 9$$

3 Write $s(t)$

$$s(t) = \frac{t^3}{3} - \frac{t^2}{2} - 6t + 9$$

4 Put in $t = 6$

$6^3 = 216$ and $6^2 = 36$:

$$s(6) = \frac{216}{3} - \frac{36}{2} - 36 + 9 = 72 - 18 - 36 + 9 = 27$$

5 Answer with units

$$s(6) = 27 \text{ cm}$$

EXAM TECHNIQUE

“Position” means $s(t)$, not the distance travelled, so you don’t need to split at the time the particle turns around. Just integrate and substitute.

Marks	Awarded for
1.5	$s(t) = \int v(t) dt.$
3	$s(t) = t^3/3 - t^2/2 - 6t + C.$
2	$C = 9$ from $s(0) = 9.$
2	Substituting $t = 6$ correctly.
1	$s(6) = 27 \text{ cm}.$

Question 4 Total: 17.5 Marks

Question 5

Least squares regression line, volumes of rotation — 17.5 marks

5(a)

[8 marks]

Find the least squares regression line $y = mx + b$, given the data points x : 0, 1, 2, 3 and y : 1, 1, 4, -1 . Recall that the slope and intercept are given by $m = \frac{N\sum(xy) - \sum x \sum y}{N\sum(x^2) - (\sum x)^2}$ and $b = \frac{\sum y - m\sum x}{N}$. Give your answers in *fractional form*. Hence sketch the line.

ANSWER

$$m = -\frac{3}{10}, \quad b = \frac{17}{10} \quad \Rightarrow \quad y = -\frac{3}{10}x + \frac{17}{10}$$

NOTE

Least squares regression isn't one of the listed MA1002 course topics. The question gives you both formulas, so it only needs careful arithmetic.

RULES YOU NEED

- Make a table with rows for x , y , xy and x^2 , then add each row. N = number of points.
- $\sum x^2$ means square each x , then add. $(\sum x)^2$ means add the x 's, then square.
- To sketch $y = mx + b$: plot the y -intercept $(0, b)$ and one more point, then join them.

WORKING — STEP BY STEP

1 Build the table of sums

Here $N = 4$:

	1	2	3	4	Sum (Σ)
x	0	1	2	3	6
y	1	1	4	-1	5
xy	0	1	8	-3	6
x²	0	1	4	9	14

2 Find m

$$m = \frac{4(6) - (6)(5)}{4(14) - 6^2} = \frac{24 - 30}{56 - 36} = \frac{-6}{20} = -\frac{3}{10}$$

3 Find b

$m\sum x = (-3/10)(6) = -18/10$, and minus a negative is a plus:

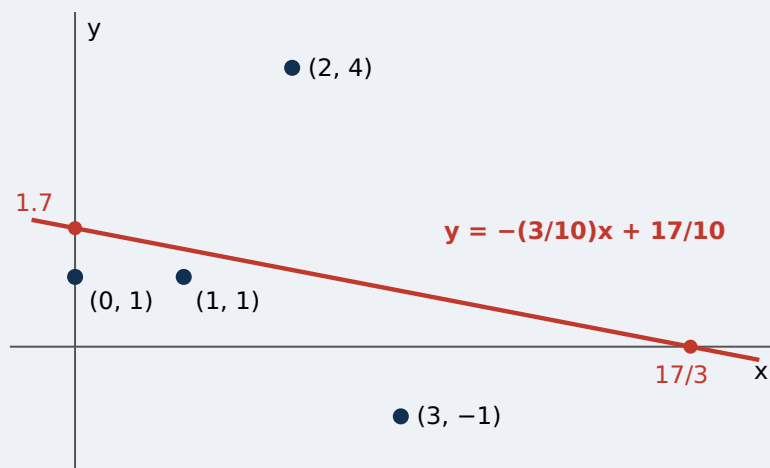
$$b = \frac{5 + 18/10}{4} = \frac{50/10 + 18/10}{4} = \frac{68/10}{4} = \frac{68}{40} = \frac{17}{10}$$

4 Write the line

$$y = -\frac{3}{10}x + \frac{17}{10}$$

5 Sketch

It cuts the y-axis at $17/10 = 1.7$. Setting $y = 0$ gives $x = 17/3 \approx 5.7$. The slope is negative, so the line goes down to the right:

**EXAM TECHNIQUE**

Keep everything as fractions (no calculator). Simplify at the end: $-6/20 = -3/10$ and $68/40 = 17/10$.

Marks	Awarded for
2	Sums $\Sigma x = 6$, $\Sigma y = 5$, $\Sigma xy = 6$, $\Sigma x^2 = 14$ (0.5 each).
2	$m = -3/10$.
2	$b = 17/10$.
2	Sketch: correct intercept and negative slope, with the data points.

5(b)**[9.5 marks]**

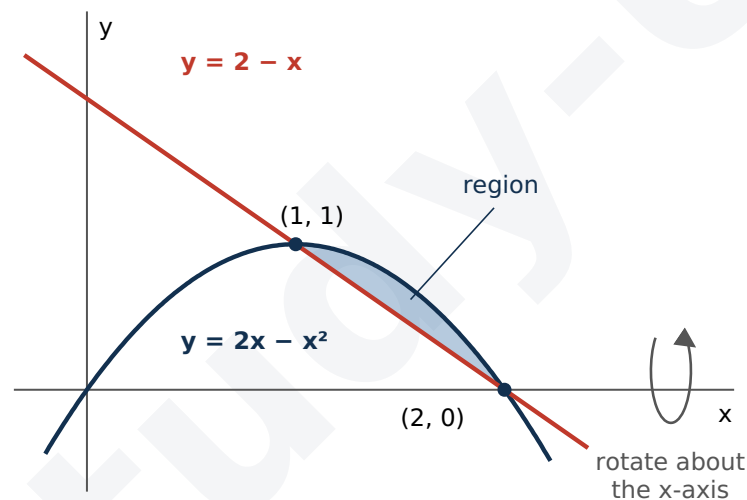
Calculate the volume generated by rotating around the x-axis the region bounded by the graphs of $y = 2x - x^2$ and $y = 2 - x$.

ANSWER

$$V = \frac{\pi}{5} \text{ cubic units}$$

RULES YOU NEED

- Volume about the x-axis: $V = \int_a^b \pi y^2 dx$.
- For a region between two curves: $V = \pi \int (y_{\text{outer}}^2 - y_{\text{inner}}^2) dx$.
- $(a - b)^2 = a^2 - 2ab + b^2$.



The small region between $x = 1$ and $x = 2$ (parabola above the line) spins around the x-axis.

WORKING — STEP BY STEP**1 Find where they meet**

Bring everything to one side and factor:

$$2x - x^2 = 2 - x \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x - 1)(x - 2) = 0 \Rightarrow x = 1 \text{ or } x = 2$$

2 Which is further from the axis?

Test $x = 1.5$:

$$2(1.5) - 1.5^2 = 3 - 2.25 = 0.75 \quad 2 - 1.5 = 0.5 \Rightarrow \text{outer: } y = 2x - x^2, \text{ inner: } y = 2 - x$$

3 Set up the volume

$$V = \pi \int_1^2 [(2x - x^2)^2 - (2 - x)^2] dx$$

4 Expand both squares

$$(2x - x^2)^2 = 4x^2 - 4x^3 + x^4 \quad (2 - x)^2 = 4 - 4x + x^2$$

$$(4x^2 - 4x^3 + x^4) - (4 - 4x + x^2) = x^4 - 4x^3 + 3x^2 + 4x - 4$$

5 Integrate

$$V = \pi \left[\frac{x^5}{5} - x^4 + x^3 + 2x^2 - 4x \right]_1^2$$

6 Put in the limits

$$x = 2: \quad \frac{32}{5} - 16 + 8 + 8 - 8 = \frac{32}{5} - 8 = -\frac{8}{5}$$

$$x = 1: \quad \frac{1}{5} - 1 + 1 + 2 - 4 = \frac{1}{5} - 2 = -\frac{9}{5}$$

$$V = \pi \left(-\frac{8}{5} - \left(-\frac{9}{5} \right) \right) = \pi \cdot \frac{1}{5} = \frac{\pi}{5}$$

EXAM TECHNIQUE

Subtract the whole of $(2 - x)^2$ in brackets: $-(4 - 4x + x^2) = -4 + 4x - x^2$. The bracket values are negative, but top – bottom is positive, as a volume must be.

Marks	Awarded for
1.5	Intersections $x = 1$ and $x = 2$.
1	Outer $y = 2x - x^2$, inner $y = 2 - x$.
2	$V = \pi \int_1^2 [(2x - x^2)^2 - (2 - x)^2] dx$.
2	Expanded: $x^4 - 4x^3 + 3x^2 + 4x - 4$.
1.5	Integrated correctly.
1.5	$V = \pi/5$.

Question 5 Total: 17.5 Marks